

Testing and Quality Assurance Plan

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Version: 1.0

Status: Draft for QA approval

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1. Quality objective

The product must be correct in four dimensions:

1. Users understand the health message.
2. Users can operate the interface.
3. Content is safe, accurate, and appropriate.
4. The application works reliably on real target devices and networks.

Visual polish alone is not quality evidence.

2. Quality gates

A release cannot proceed with an unresolved P0 or P1 issue in:

- Clinical safety.
- Privacy or security.
- Core navigation.
- Accessibility.
- Translation of critical instructions.
- Core device performance.
- Publication integrity.

3. Test layers

3.1 Schema and unit tests

Test:

- Publication schema.
- Lesson schema.
- Locale schema.
- Timeline events.

- Visual target references.
- Content review states.
- Reader state transitions.
- Permission functions.
- SVG sanitization.
- Local progress.

3.2 Integration tests

Test:

- Public publication API.
- Locale resolution.
- Approved-content filtering.
- Audio and transcript integration.
- Visual target mapping.
- Review workflow.
- Publication versioning.
- Local cache behaviour.

3.3 End-to-end tests

Test complete journeys:

- Open publication.
- Select language.
- Start lesson.
- Use linear mode.
- Use magazine mode.
- Turn pages.
- Play narration.
- Open transcript.
- Use visual target.
- Switch language.
- Disable motion.
- Use offline lesson.
- Recover from missing asset.

3.4 Accessibility tests

- Keyboard-only operation.
- Screen-reader order.
- Focus after page change.
- Text scaling.
- Contrast.
- Reduced motion.
- Captions and transcript.
- No colour-only meaning.
- Touch target reachability.

3.5 Visual regression tests

Capture:

- Cover.
- Mobile lesson.
- Desktop spread.
- Long translated title.
- Missing visual.
- Audio loading.
- Reduced-motion mode.
- Lite profile.
- Facilitator mode.

3.6 Performance tests

Test:

- Cold start.
- Warm start.
- Slow network.
- Low-end Android.
- Long publication.
- Repeated page turns.
- Audio loading.
- WebGL disabled.
- Memory after repeated navigation.

4. Comprehension testing

4.1 Objective

Measure whether users understand and can act on the message, not whether they admire the animation.

4.2 Method

- Recruit representative users.
- Test each prototype lesson.
- Ask users to explain the message in their own words.
- Ask what they would do next.
- Ask what the illustration means.
- Observe confusion without correcting immediately.
- Record language and accessibility observations.
- Revise and retest.

4.3 Proposed target

At least 80% of representative pilot users should correctly explain the key message and identify an approved safe next action. This is a proposed target requiring product-owner confirmation.

5. Device test matrix

Create a real-device matrix covering:

- Low-cost Android phone.
- Mid-range Android phone.
- Older iPhone.
- Modern iPhone.
- Desktop Chrome.
- Desktop Firefox.
- Desktop Safari.
- Tablet.

Record:

- Device and OS.
- Browser version.
- Network profile.
- Rendering profile.
- First content time.

- Audio start time.
- Page-turn stability.
- Memory observation.
- Errors.

6. Network test matrix

- Offline.
- Intermittent connection.
- Slow 3G.
- Fast 4G.
- Wi-Fi.
- Asset request failure.
- API timeout.
- Partial cache.

7. Audio tests

- Audio starts.
- Audio pauses.
- Audio resumes.
- Audio seeks.
- Audio speed changes.
- Transcript remains synchronized.
- Visual target remains synchronized.
- Audio missing fallback works.
- Locale-specific timing works.

8. Reader physics tests

- Page opens.
- Next and previous work.
- Page order is correct.
- Keyboard works.
- Touch swipe works.
- Drag commits.
- Drag cancels.
- Fast swipe does not skip unexpectedly.
- Focus moves correctly.

- Reduced motion removes heavy movement.
- WebGL failure falls back safely.

9. Content validation tests

- Required source exists.
- Clinical status is approved.
- Translation status is approved.
- Accessibility description exists.
- Asset license exists.
- Review date exists.
- Timeline target exists.
- Transcript token exists.
- Publication version is valid.

10. Security tests

- Anonymous draft access denied.
- Withdrawn content denied.
- Learner data isolation works.
- Editor cannot self-approve.
- Unsafe SVG rejected.
- Protected mutation rejects missing role.
- Audit event is written.
- Secrets do not appear in build output.

11. Defect severity

P0: Safety, privacy, unauthorized access, or major data loss.
P1: Core learning or navigation unusable.
P2: Significant accessibility, translation, or performance defect.
P3: Cosmetic or low-impact defect.

12. Release scorecard

Clinical accuracy: Pass / Fail / Pending
Comprehension: Pass / Fail / Pending
Translation: Pass / Fail / Pending
Accessibility: Pass / Fail / Pending
Security: Pass / Fail / Pending
Performance: Pass / Fail / Pending
Offline: Pass / Fail / Pending
Asset rights: Pass / Fail / Pending

Rollback: Pass / Fail / Pending
Documentation: Pass / Fail / Pending

No release is green while a required item is Pending.

13. Evidence package

Store:

- CI output.
- Test reports.
- Screenshots.
- Browser recordings.
- Device results.
- Clinical review records.
- Translation sign-off.
- Accessibility audit.
- Security review.
- Performance report.
- Release manifest.

14. QA acceptance criteria

- Automated tests pass.
- Manual high-risk paths pass.
- Representative user testing is complete for the milestone.
- All release-critical review statuses are approved.
- No unresolved P0/P1 defect remains.
- Project memory contains the evidence and limitations.